

IMPERIAL GLOBAL
SINGAPORE



IN-CYPHER Hackathon

Network: CREATE_GUEST

Day 1

Day 1 Schedule

Time	Participant programme
10:30–11:30	Welcome Workshop CRPO
11:30–13:00	HTX booth and Build your agent
13:00–14:00	Lunch
14:00–15:15	Build your agent
15:15–16:15	Networking and Afternoon tea at Royals Bistro (No Coding)
16:15–17:15	Build your agent
17:15–17:30	Pack up

Fire Drill & Emergency Evacuation Procedures

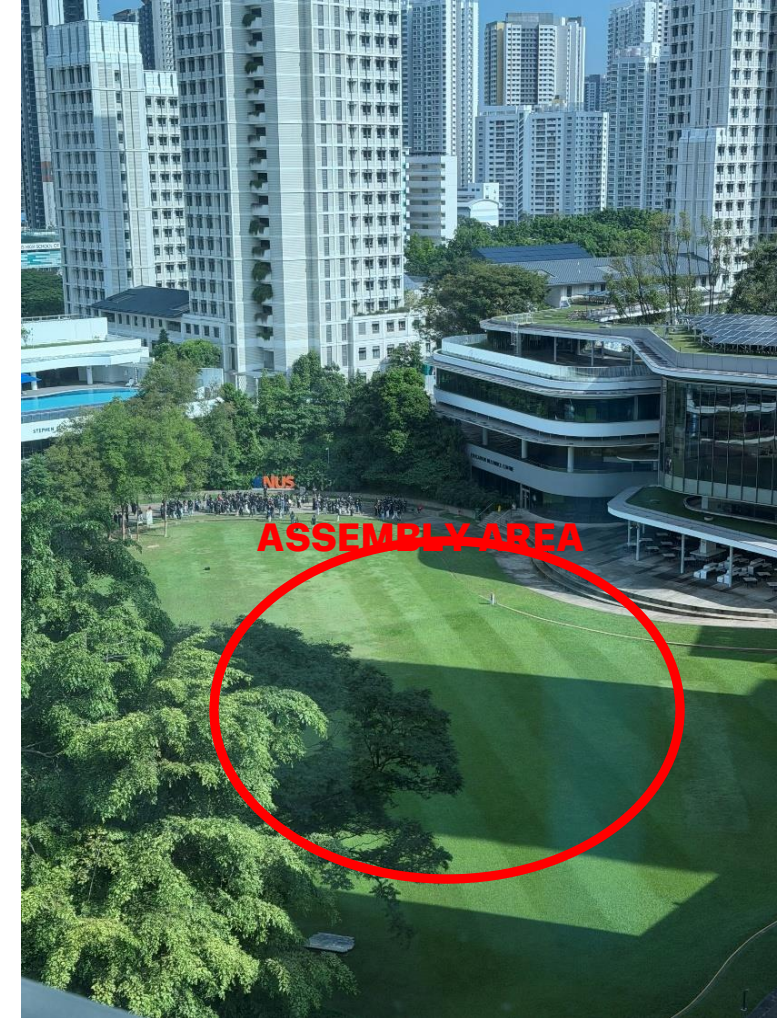
Two-Stage Alarm Procedure

First Stage Alarm

- Stop all activities
- Alert surrounding people
- Get ready for evacuation
- EM Lock doors will be released
- Automated investigation message will be broadcasted through PA system

Second Alarm (Continuous ringing):

- Start evacuation
- Assemble at the Assembly Area (on the grass patch outside Starbucks) and report attendance to the Fire Warden
- Assemble according to your parent organization (IGS)
- Do NOT re-enter the building until further notice



Fire Drill & Emergency Evacuation Procedures

Things to Note



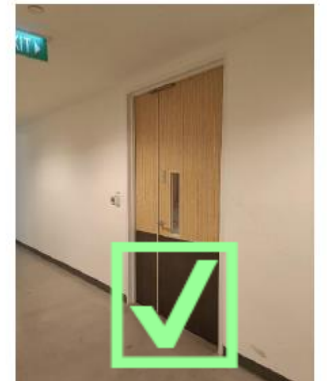
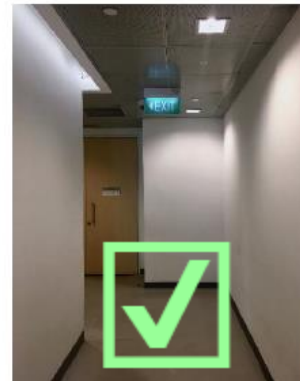
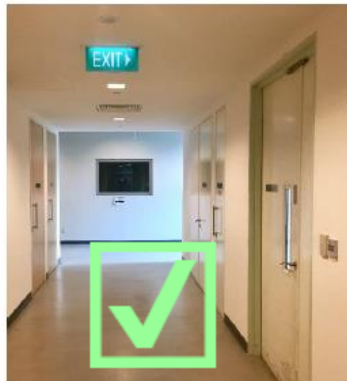
DO NOT USE THE MAIN STAIRS



DO NOT TAKE THE LIFTS



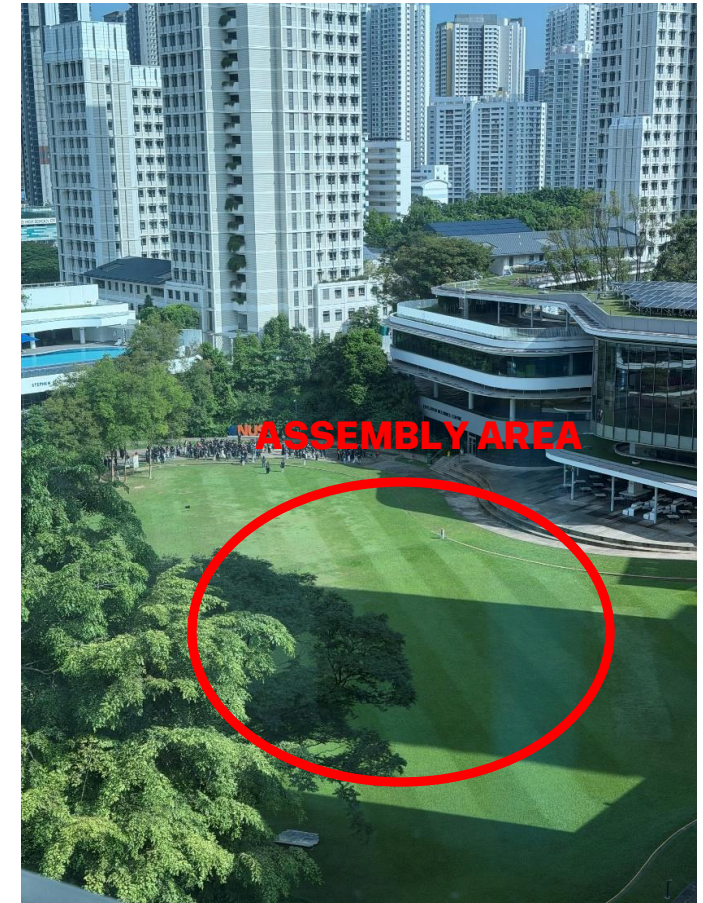
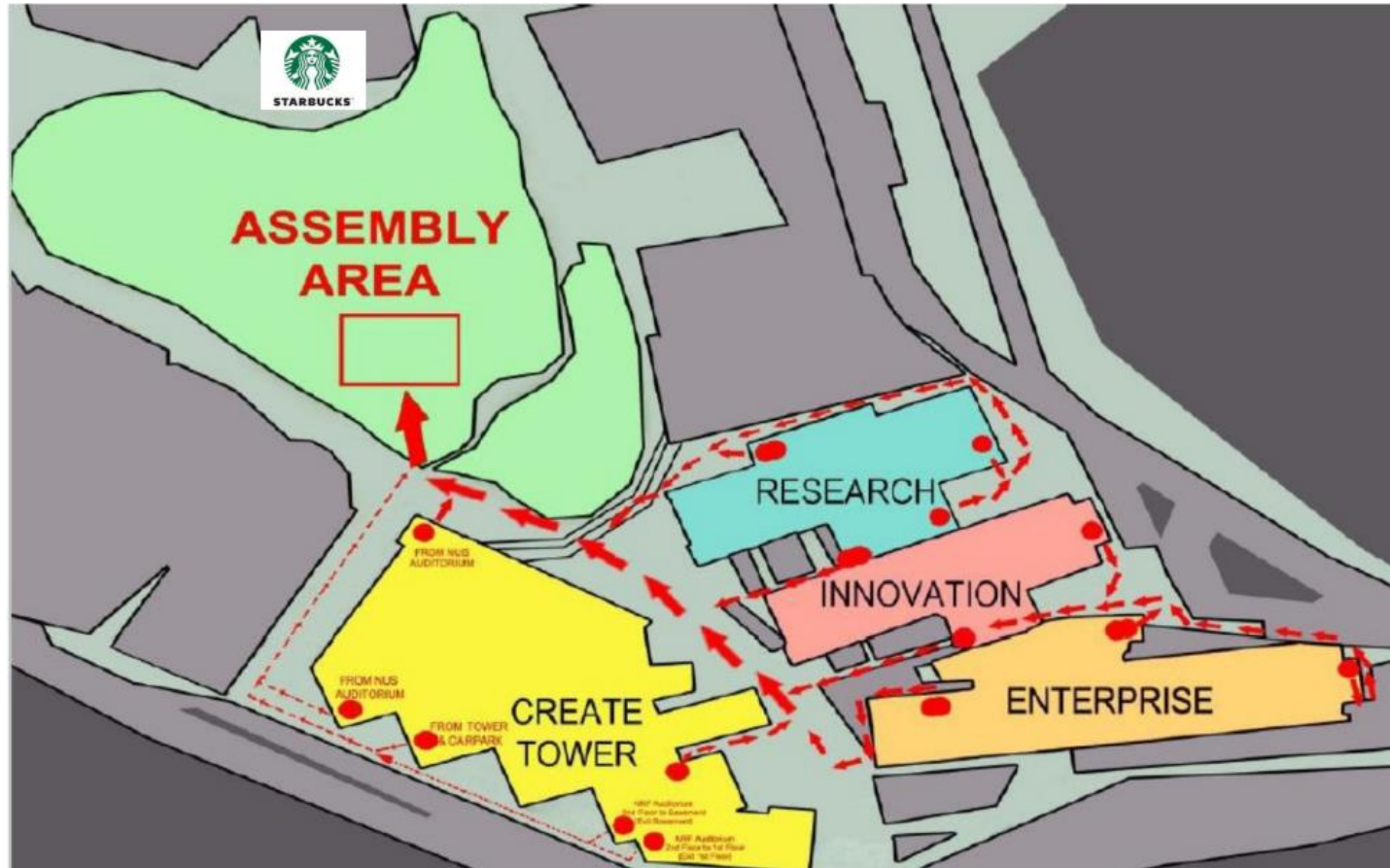
Building Evacuation through nearest Exit Doors



Follow the EXIT sign to smoke stop lobby and then take the stairs down to level 1

Fire Drill & Emergency Evacuation Procedures

Where to Assemble

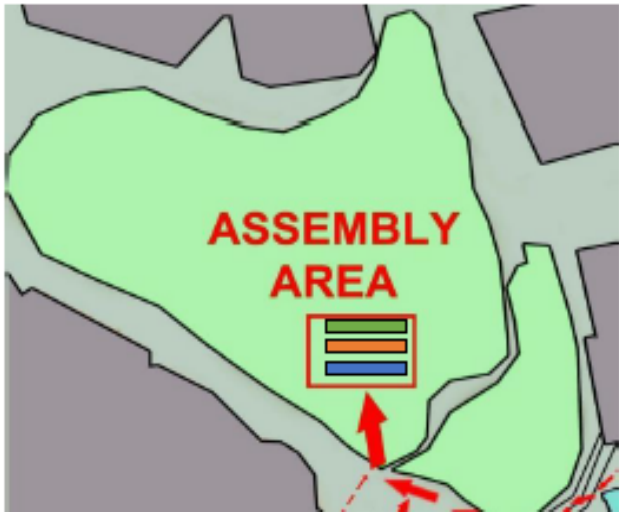


Fire Drill & Emergency Evacuation Procedures

How to Assemble

Assemble according to your parent organization (IGS)

Gathering at the Assembly Area



SMART	SHARE	NGRIP	TUM	SJTU	CARES		
CNRS	IGS	CLIC	SEC (ETH)	BEARS	NUS FOS	SingFar m	AI4S
S&TPPO	NRF	TC	Data Center	NUS MSE	IARCS		

Workshop

IN-CYPHER HACKATHON

The Autonomous Agent Challenge

Competition rules and some basic understanding of AI agent

Presenter: Xie Yixuan

The Overall

Part 1

The rules

Your budget, why the agent must never stop, how sub-agents and dynamic challenges work, and how you're scored.

Part 2

Foundations

What an AI agent is, how it runs, the tools it uses, and the three things that make one strong: skills, memory, harness.

PART 1 · THE RULES

Build one fully autonomous agent that scores as many points as it can.

How to make your agent to get more scores for you without interruption.

Four Important Rules

R1 • Budget

\$100 of tokens per team

When it's spent, your run ends. That's your entire compute allowance.

R2 • Autonomy

Hands-off once it starts

Run it unattended. Stepping in to help will make you lose points.

R3 • Parallelism

No two dynamic problems

Run many sub-agents at once is permitted, but never two dynamic challenges simultaneously.

R4 • Scoring

Points, then speed

Your score is the sum of challenge points. Ties are broken by finish time.

RULE 1 • BUDGET

A fixed \$100 in tokens

Every team gets the same allowance: **\$100** of model usage. When it runs out, your session stops — solved or not.

There are no top-ups. So efficiency is part of the game: how to choose a model for the brain and how to save the token.

YOUR ALLOWANCE

\$100

per team • one-time • no refills

Spend it well. Expensive model ends early, avoid re-reading the same output, let cheaper steps do routine work.

RULE 2 - AUTONOMY

Run it hands-off

From launch until the budget is gone, the agent runs on its own. **No human may step in** to unstick it, restart it, or hand it hints.

- 1: You can add some logs to help me to track your agent's progress and (if necessary) make some improvements
- 2: Once you ask us to step in to restart your agent, 100 scores will be penalized for one time.

PENALTY

Stepping in costs points.

Manual intervention is penalized.
(But your agent can restart itself)

WHY THIS RULE

The contest measures autonomy.

How to keep it running

01 · Timeouts everywhere

Put a time limit on every tool call. A hung connection should fail fast, not freeze the agent.

02 · Catch and continue

Wrap each step so an error is logged and the loop moves on — never let one crash end the run.

03 · A watchdog

Run a supervisor that watches for silence and restarts the loop if the agent goes quiet.

04 · Never ask a human

No interactive prompts. When stuck, pick a default or switch challenge — always have a next move.

RULE 3 · PARALLELISM

Don't run two dynamic together

Your agent may create some sub-agents to solve a questions to raise the efficiency.

The single restriction: **no two dynamic challenges at once**. Static challenges have no such limit — parallelize them freely.



Allowed: five sub-agents on five static challenges in parallel.



Allowed: one dynamic challenge while static ones run alongside.



Not allowed: two dynamic challenges live at the same moment.



Static vs. dynamic challenges

Static

A fixed artifact — a file to download and analyze: a binary, a packet capture, a ciphertext.

Dynamic

A live service spun up just for your team — a web app, a host, a pwn target you connect to.

Design implication: build a queue for dynamic challenges — start one, solve or give up, tear it down, then start the next.

RULE 4 • SCORING

Points first, time second

Your score is simply the **sum of the point values** of the challenges your agent solves. Harder challenges are worth more.

Time is only a tiebreaker: if two teams finish on the same score, the one that reached it faster ranks higher.

1

Total points

Add up every solved challenge's value.

2

Finish time

Applied only when points are equal.

Note: don't chase speed at the cost of a solve — one more challenge always beats a faster tie.

How to score higher

Team	Solved	Points	Finished	Rank
Team C	5 challenges	1,100	3 h 25 m	1st
Team B	5 challenges	1,100	4 h 10 m	2nd
Team A	4 challenges	950	3 h 40 m	3rd

C beats B — same 1,100 points, but C got there 45 minutes sooner.

A loses to both — faster than B, but fewer points. Points come first.

Three important reminders

\$20 tryout

A separate \$20 allowance to practice and test your setup before the scored run — it does not touch your \$100.

Any model you like

Use whatever model you prefer; it is entirely your choice. Your \$100 covers the usage.

Keep a log (suggested)

We suggest a simple logging function, so you can trace your agent's progress and keep improving as you go.

What is an AI agent?

A CHATBOT

Question



Model



Reply

Then it stops. It can't run anything or check if it was right.

AN AGENT

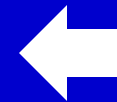
Goal



Think



Act

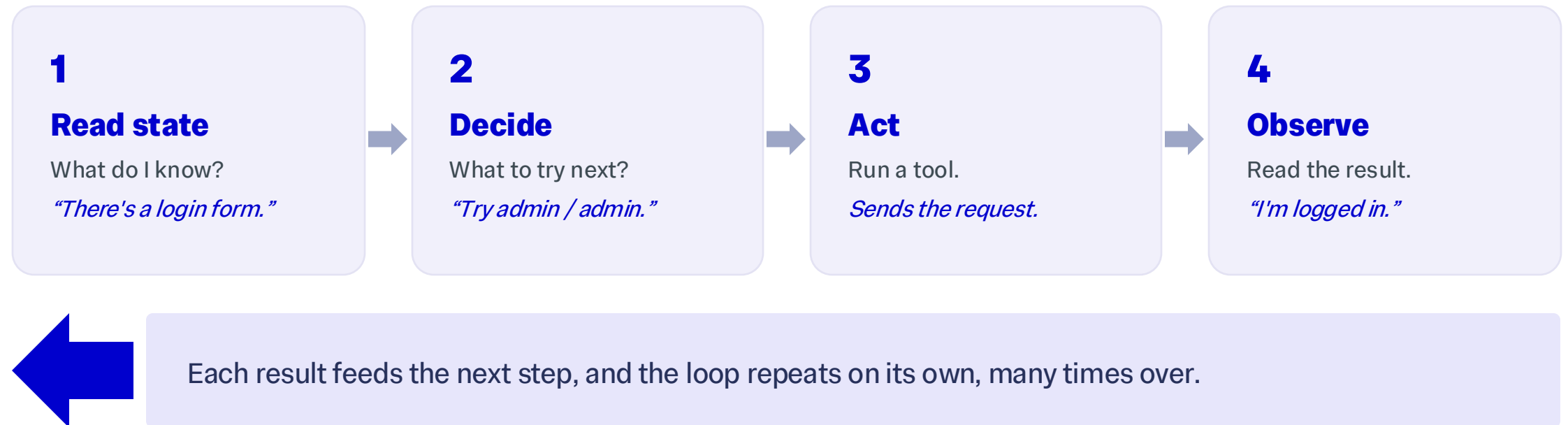


observe, then go again — on the live target

Chatbot only answers; An agent takes real actions on the target to reach the goal.

How it runs on its own

The general loop for the AI agent



/t repeats look, plan, try, check automatically — much faster than by hand.

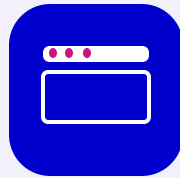
What the agent can actually do

A tool is any action the model is allowed to take. For CTF, these five carry most of the work.



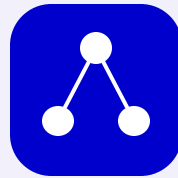
Shell

Run any command in a terminal.



Web

Open pages, send HTTP requests.



Network

Connect to a host, scan ports.



Files

Read a binary, a capture, a key.

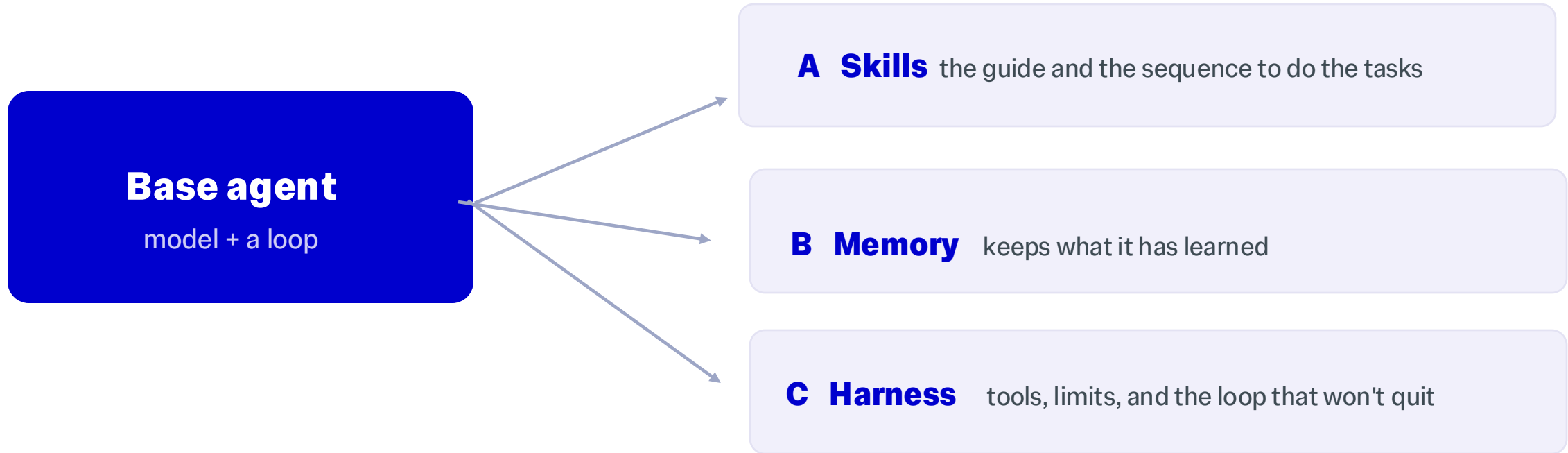


Submit

Send a flag to score it.

The agent can do almost everything like a smart human if you give it enough "knowledge"

Three things make an agent strong



skills are the playbooks you give them, memory is the notebook they keep, the harness is the desk, tools and rules around them.

Skills

A skill is to tell the AI agent to do things in specific steps

SKILL - web recon

- 1 check robots.txt
- 2 probe common admin paths
- 3 try default credentials
- 4 only then, get creative

Without it

guesses, tries random things, burns budget.

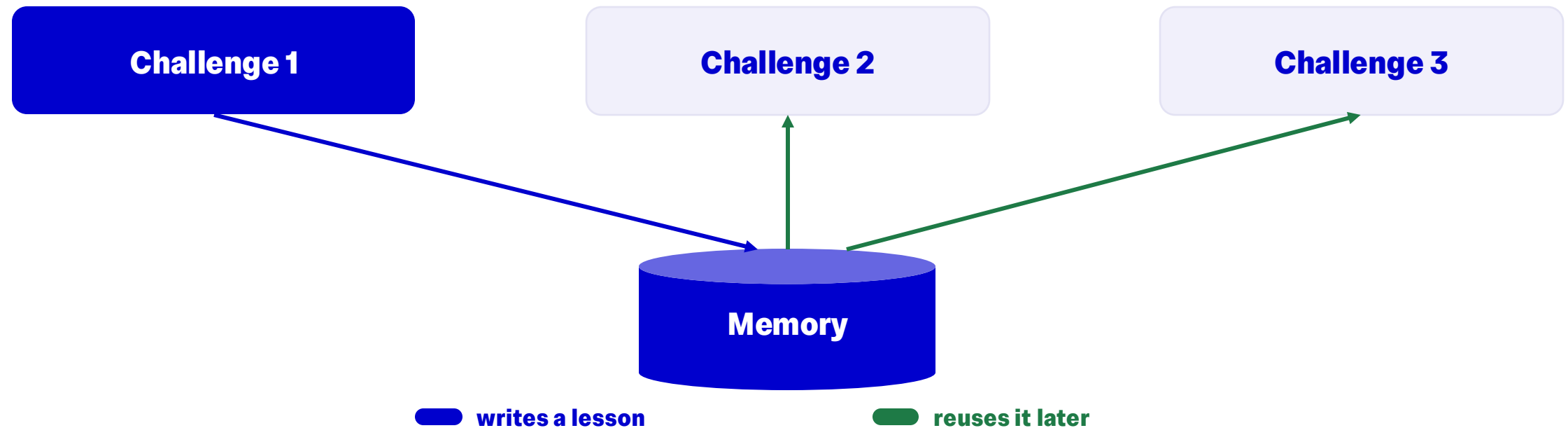
With it

runs the moves that work, then thinks — fewer wasted steps.

prepare one skill per category — web, crypto, pwn, forensics — with the opening moves a strong player always makes.

Memory — lessons that carry over

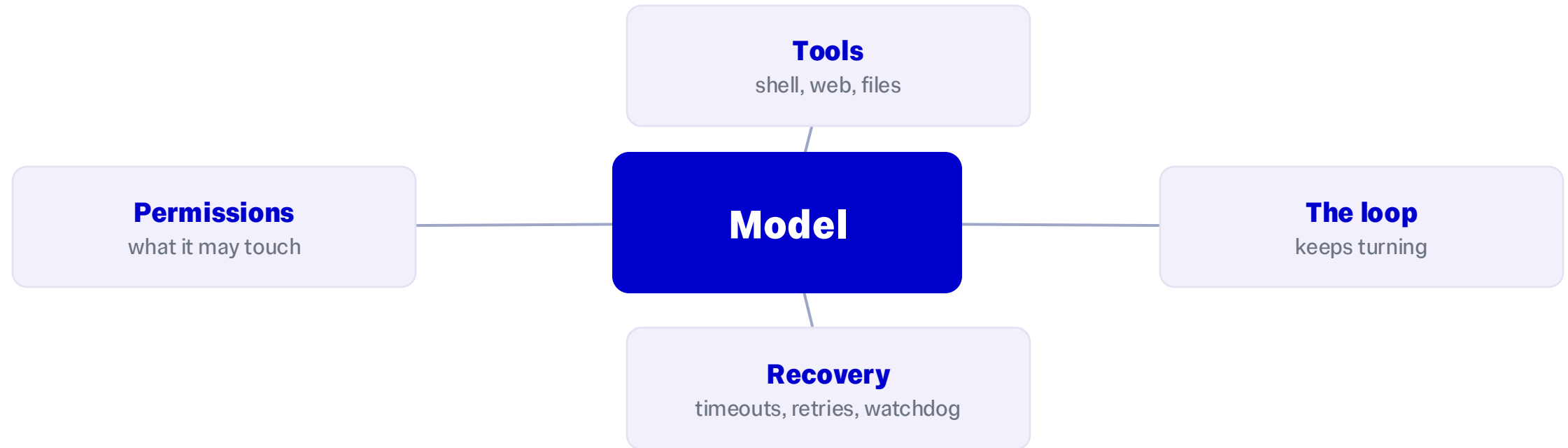
A model forgets when a task ends. Memory lets one lesson pay off on the next challenge.



If this time: “the flag hides in an environment variable” — saved , then reused on every similar challenge.

Harness — what keeps it alive

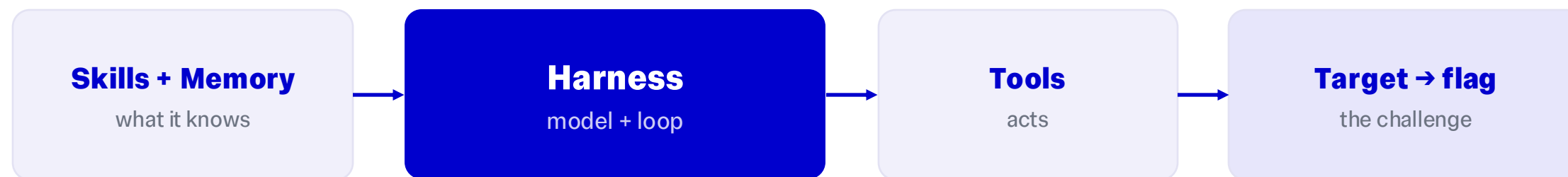
Everything wrapped around the model — tools, limits, the loop, and error recovery.



lets the agent recover from an error and carry on, instead of stopping at the first one.

What a full agent looks like

Loop, tools, skills, memory, a tough harness — it recons, exploits, submits, moves on.



```
agent.log
```

```
[recon] boot instance → scan ports  
[skill] web: try default creds → ok  
[act] read /api/secret → parse  
[memory] note: env-var flag pattern  
[submit] INCYPHER{...} → correct  
[loop] next challenge ...
```

What's inside the Docker image

18 layers · ~1 GB

The agent is one container. Here's everything inside — grouped by what each part does.

■ **Debian base (trixie)** — the OS everything runs on

87 MB

■ **Python 3.12 runtime** — the language the agent is written in

~46 MB

■ **System tools · apt** — curl, netcat, gdb and friends

483 MB

■ **Python libraries · pip** — pwntools, pycryptodome, requests

362 MB

■ **The agent · /opt/agent** — the loop: run_bash + submit_flag

33 KB

■ **Runtime config** — env vars + start command

0 B

Do everything with the help of AI

Claude Code is an agent that runs in your terminal.

You can do anything with the help of it.

1 Install it
Claude -install
`curl -fsSL https://claude.ai/install.sh | bash`

2 Start it
run `claude` in your folder

3 Give it a goal
type what you want, in plain English

terminal

`$ claude`

`> find the flag on http://target:8000`

- reads the page, tries admin/admin
- logged in → GET /api/secret
- parse response → found the flag

`submit INCYPHER{...} ✓ solved`

Preparation checklist

- ✓ Use our reference agent for a head start, or build your own from scratch.
- ✓ Improve it — timeouts, catch-and-continue, a watchdog, so it keeps working.
- ✓ Add some limits and skills to your agents.
- ✓ Spend the \$100 budget deliberately — set give-up rules so it isn't wasted on dead ends.
- ✓ Test a full unattended run before the day and add your logs — can try to roughly run today.

Questions?

Build it to run unattended, use the budget well, and let it work. Good luck.



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CRPO Translation Internship Programme (Nationwide)



MISSION

The CRPO's CyberSG Translation Internship Programme aims to strengthen Singapore's cybersecurity ecosystem and talent pipeline by providing students with hands-on experience on CRPO-funded research and innovation projects.



WHO CAN APPLY

Students from Singapore Autonomous Universities and polytechnics, including undergraduates, Master's and PhD candidates.



RELEVANT DISCIPLINES



Cybersecurity



Computing/AI



Engineering



Business



Economics



**Public Policy
& others**

Technical Roles Example

- Frontend Development
- Backend Development
- LLM Supply Chain Security
- Technical Demo Development

Business Roles Example

- Product Management
- Sales & Business Development
- Marketing
- Security Business Product Analysis

Hiring Summary to Date

45

Total Interns Hired

17

Ongoing Interns

±15

Interns to be recruited

Current Opening Roles

- **NUS & i-Sprint** — Technical Demo Development Intern
- **SMU & digiDations** — AI Multimedia & Content Creation Intern

Post-Internship Programme

- **A*STAR & StrongKeep** — Intern is hired as full-time staff
- **NTU & PQStation** — Intern is now a Co-founder in A*Star Spinoff

New Internship Posting Cycle: End 2026/Early 2027

Where to Find Our New Internship Opportunities?

University or Polytechnic's Official
Internship Application Portals

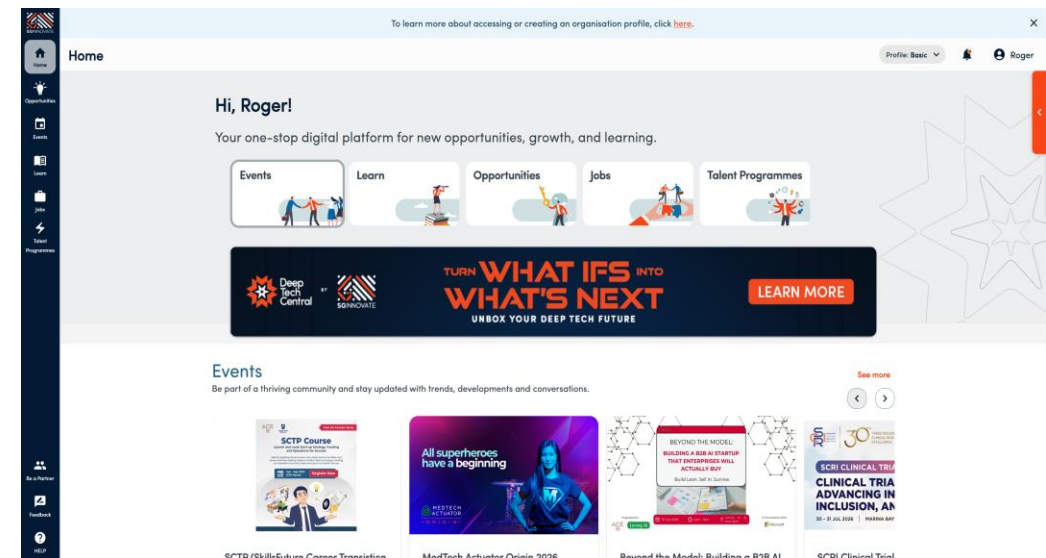


CRPO works closely with career offices across
various schools.

POLYTECHNICS
NP · SP · RP · NYP · TP

UNIVERSITIES
NTU · NUS · SMU · SIT · SUTD

SGInnovate
Deep Tech Central



Sample Projects, Institutions, and Collaborators

Project Team	Collaborator
SUTD - Cloud-based Intelligent OT-Centric Asset Monitoring for Critical Infrastructure	Ensign & PUB
NTU - AutoSOC: A Knowledge-Driven Agentic AI System for Automated SOC Operations	Antarex
NUS - A Closed-Loop, Interpretable, and Lightweight AI-based IAM Framework	i-Sprint
NUS - Bridging the AI trust gap with LLM-powered Program Analysis	AIQURIS
IARCS - Intelligent and Accessible Threat Modeling	Cybernatics, Critical Systems Labs & OverArX
ETH - FIREx Future Internet Resilience Exchange	Nexusguard
SMU & NTU - AutoIntelligence: An End-to-End Agentic Platform for Software Security Intelligence	Digidations
NTU - ComplianceAgent: Autonomous Cybersecurity Compliance through Multi-Agent Coordination	Cyber Sierra
NTU - Cyber-Physical Risk Assessment for Power Grid Networks using Advanced Digital Twin	HY Mechanical and Electrical (M&E) Engineering Consultancy
NTU - Efficient Fully-Homomorphic Encryption-Protected Machine Learning Models	Zama
NUS - Agentic-VAPT: Empowering Vulnerability Assessment and Penetration Testing using Agentic AI	Ensign
A*STAR - vCISO: An Agentic Approach to Build a Virtual CISO	StrongKeep
NTU – Migration to Quantum Safe Systems (MIQS)	PQStation
SMU – From Risk Identification to Risk Management: A Systematic Approach to Mitigating LLM Supply Chain Risks. SCIS	AgentSure
NTU – From Past to Present: Analyzing Security Vulnerabilities of 3GPP Specifications	Protofy Technology Pte. Ltd.
A*STAR – Telecom Cybersecurity Innovation Centre (TCIC Phase 1). I2R	ST Engineering
NTU & IHPC – Cyber Sentinel: A Unified Framework for Safeguarding Foundation Models	AI Guardian
NUS & NTU & SMU – Secure, Private, and Verified Data Sharing for Large Model Training and Deployment	Dataillusion

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Pack Up

View and download your photos now



Day 2

Day 2 Schedule

Time	Participant programme
09:30–10:00	Check-in and Welcome
10:00–12:30	Agent run in progress
12:30–13:30	Lunch (Agent keeps running)
13:30–15:00	Agent run in progress
15:00–15:30	Afternoon tea (Agent keeps running)
15:30–16:30	Agent run in progress (ends 16:30 sharp)
16:30–17:00	Announcement of winners

View and download your photos now



Agent run in progress



Agent run in progress



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Imperial Global Singapore IN-CYPHER Hackathon 2026

Agent run in progress

End



Winners

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SINGAPORE



Thank you for coming!